
Week 37: Long-Run Behavior of Markov Chains (Part 2)

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

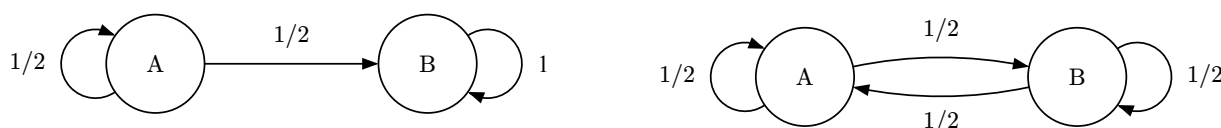
1. Theory

Last week we focused on *regularity* of Markov chains and saw that this property is useful for proving convergence. However, regularity can fail because of periodicity, reducibility, or transient states. We also want to extend our analysis to infinite state spaces.

Definition 1.1. (*Communication*) Let $\{X_n : n = 0, 1, \dots\}$ be a Markov chain with transition probability matrix $\mathbf{P}$.

- State j is *accessible* from state i if there exists an integer $n \geq 0$ such that $P_{i,j}^{(n)} > 0$.
- If states i and j are accessible from each other, they *communicate*. We write this as $i \sim j$.

Example 1.2. In the first scenario, B is accessible from A, but A is not accessible from B. Therefore $A \not\sim B$. In the second scenario, A and B are accessible from each other, so $A \sim B$.



Theorem 1.3. (*Communication is an equivalence relation*) *Communication is an equivalence relation:*

- *Reflexivity:* $i \sim i$ for every state i .
 - This follows since $P_{i,i}^{(0)} = 1$.
- *Symmetry:* if $i \sim j$, then $j \sim i$.
- *Transitivity:* if $i \sim j$ and $j \sim k$, then $i \sim k$.
 - This is since if $P_{i,j}^{(n)} > 0$ and $P_{j,k}^{(m)} > 0$, then $P_{i,k}^{(n+m)} = \sum_{l=0}^{\infty} P_{i,l}^{(n)} P_{l,k}^{(m)} \geq P_{i,j}^{(n)} P_{j,k}^{(m)} > 0$.

It therefore partitions the state space into equivalence classes of communicating states.

Example 1.4. Consider the Markov chain with transition probability matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

The equivalence classes for both $\mathbf{A}$ and $\mathbf{B}$ are $\{0, 1\}$ and $\{2, 3\}$.

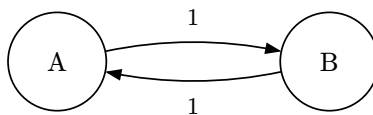
Definition 1.5. (*Irreducibility*) A Markov chain is *irreducible* if it has exactly one communication class, meaning that all states communicate. Otherwise, the chain is *reducible*.

Definition 1.6. (*Periodicity*) The *period* of state i , denoted by $d(i)$, is

$$d(i) = \gcd\{n \geq 1 : P_{i,i}^{(n)} > 0\}.$$

If $P_{i,i}^{(n)} = 0$ for every $n \geq 1$, we define $d(i) = 0$. A state with $d(i) = 1$ is called *aperiodic*. See that $d(i) = 1$ if $P_{i,i} > 0$.

Example 1.7. Consider the following Markov chain.



Here we can see that

$$P_{i,i}^{(n)} = \begin{cases} 1 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}.$$

Hence, $d(i) = \gcd\{2, 4, 6, \dots\} = 2$.

Theorem 1.8. If $i \sim j$, then $d(i) = d(j)$. Thus, periodicity is a property of a communication class. In particular, for an irreducible chain it makes sense to call the entire chain periodic or aperiodic.

Definition 1.9. (First-return probabilities)

$$f_{i,i}^{(n)} = P(X_n = i, X_\nu \neq i \text{ for } \nu = 1, 2, \dots, n-1 \mid X_0 = i), \quad n \geq 1,$$

and we set $f_{i,i}^{(0)} = 0$. The probability of ever returning to state i is $f_{i,i} = \sum_{n=1}^{\infty} f_{i,i}^{(n)}$.

Definition 1.10. (Recurrent and transient states) State i is recurrent if $f_{i,i} = 1$. It is transient if $f_{i,i} < 1$.

Intuition:

- Starting from a recurrent state, that state is visited infinitely often with probability 1. A transient state is visited only finitely often with probability 1.
- If a state is recurrent, the probability of returning to it after some finite length of time is 1. If it is transient, the probability of returning to it after some finite length of time is less than 1.

Theorem 1.11. Recurrence and transience are class properties. If $i \sim j$ and i is recurrent, then j is recurrent.

Theorem 1.12. A state i is recurrent if and only if $\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty$.

Definition 1.13. (Mean recurrence time) The mean recurrence time of state i is

$$m_i = \sum_{n=1}^{\infty} n f_{i,i}^{(n)},$$

the expected time between successive visits to state i .

Definition 1.14. (Positive and null recurrence) A recurrent state i is positive recurrent if $m_i < \infty$, null recurrent if $m_i = \infty$.

Positive recurrence and null recurrence are class properties.

Definition 1.15. (Stationary distribution) A probability vector $\pi = (\pi_0, \pi_1, \dots)^\top$ is a stationary distribution if

$$\sum_{i=0}^{\infty} \pi_i = 1 \quad \text{and} \quad \pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j}, \quad j = 0, 1, \dots$$

For finite state spaces, $\pi = \mathbf{P}^\top \pi$ and $\mathbf{1}^\top \pi = 1$.

Theorem 1.16. (Existence and uniqueness of stationary distributions) An irreducible Markov chain has a stationary distribution if and only if it is positive recurrent.

In that case, the stationary distribution is unique and $\pi_i = 1/m_i$.

Theorem 1.17. (When the stationary distribution is limiting) An irreducible, positive recurrent, and aperiodic Markov chain has a (unique) limiting distribution.

It is equal to the unique stationary distribution

$$\lim_{n \rightarrow \infty} P_{j,i}^{(n)} = \pi_i = \frac{1}{m_i}$$

for all states i, j .

Corollary 1.18. *(Finite irreducible chains) Every finite irreducible Markov chain is positive recurrent, and therefore has a unique stationary distribution. If it is also aperiodic, this stationary distribution is also the limiting distribution.*

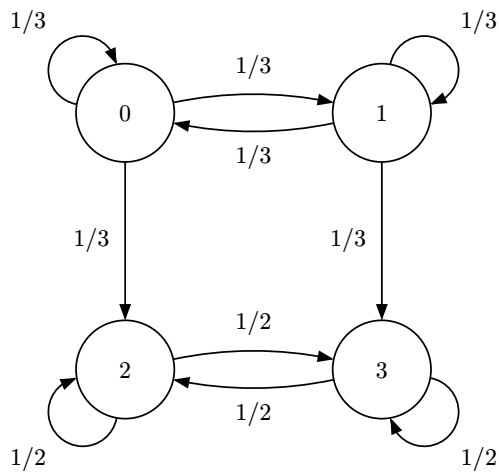
Problem 1.1. Consider two Markov chains with transition probability matrices

$$\mathbf{A} = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

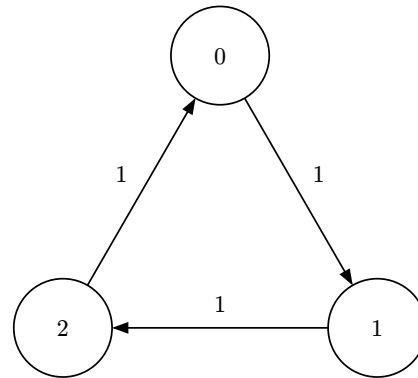
Determine for each Markov chain how many equivalence classes it has and whether it is irreducible or reducible.

Solution.

Drawing the transition diagram, we see



(a) Probability transition diagram for $\mathbf{A}$



(b) Probability transition diagram for $\mathbf{B}$

For $\mathbf{A}$, it is clear that $0 \sim 1$ and $2 \sim 3$, but 0 and 1 do not communicate with 2 and 3. Therefore, there are two equivalence classes: $\{0, 1\}$ and $\{2, 3\}$. As we have more than one equivalence class, the Markov chain is reducible.

For $\mathbf{B}$, we have $0 \sim 1$, $1 \sim 2$, and $2 \sim 0$. Therefore, all states communicate with each other, and there is only one equivalence class $\{0, 1, 2\}$ and the Markov chain is irreducible.

Problem 1.2. Consider the Markov chains specified by the transition probability matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1/2 & 0 & 1/2 & 0 \end{bmatrix}.$$

Assume $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$.

For each chain do the following:

- Is the Markov chain reducible or irreducible? If it is reducible, specify its equivalence classes.
- Calculate the period of each state.

Solution.

- For $\mathbf{A}$, we have $i \sim i$ for all states i , but no other states communicate. Therefore the equivalence classes are $\{0\}$, $\{1\}$, $\{2\}$, dots, $\{n-1\}$ and the Markov chain is reducible. We have $d(i) = 1$ for all states i since $P_{i,i} = 1$.
- For $\mathbf{B}$, we have $0 \sim 1$, $1 \sim 2$, dots, $\{n-2\} \sim \{n-1\}$, and $\{n-1\} \sim 0$. Therefore all states communicate with each other, and there is only one equivalence class $\{0, 1, \dots, n-1\}$ and the Markov chain is irreducible. We have $d(i) = n$ for all states i since we can only return to state i after n steps.
- For $\mathbf{C}$, we can see that all states communicate with each other, and there is only one equivalence class $\{0, 1, 2, 3\}$ and the Markov chain is irreducible. Considering state 0, the set of integers $n \geq 1$ where $P_{0,0}^{(n)} > 0$ is $\{4, 6, 8, \dots\}$, and therefore $d(0) = 2$. By [Theorem 1.8](#), we have $d(i) = 2$ for all states i .

Problem 1.3. A two-state Markov chain has the transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}.$$

Determine the first-return distribution

$$f_{0,0}^{(n)} = P(X_n = 0, X_\nu \neq 0 \text{ for } \nu = 1, 2, \dots, n-1 \mid X_0 = 0), \quad n \geq 1,$$

Solution. We have

$$f_{0,0}^{(1)} = P(X_1 = 0 \mid X_0 = 0) = 1 - a.$$

For $n \geq 2$, to return to 0 for the first time at step n , the chain must follow

$$0 \rightarrow 1 \rightarrow 1 \rightarrow \dots \rightarrow 1 \rightarrow 0.$$

By the chain rule and the Markov property,

$$f_{0,0}^{(n)} = a(1-b)^{n-2}b, \quad n \geq 2.$$

Problem 1.4. Let $\{X_n : n = 0, 1, \dots\}$ be a Markov chain with finite state space $\{0, 1, \dots, N\}$. Show that if the Markov chain is aperiodic and irreducible, then it is regular and recurrent.

Solution. Fix a state i . Since the finite chain is irreducible, from every state j there is a path to i with positive probability. Because the state space is finite, there are an integer M and a number $\varepsilon > 0$ such that, from any state, the probability of hitting i within the next M steps is at least ε . By the Markov property, the probability of avoiding i for kM steps is at most $(1 - \varepsilon)^k$, which tends to 0. Starting from i , the chain therefore returns to i with probability 1, so every state is recurrent.

Next we show that irreducibility and aperiodicity imply regularity. A regular Markov chain is one for which there exists a positive integer n such that $\mathbf{P}^n$ has strictly positive entries. Observe that:

- Since the chain is irreducible, all states communicate and for each pair of states i and j there exists a positive integer $n_{i,j}$ such that $P_{i,j}^{(n_{i,j})} > 0$.
- Since the chain is aperiodic, for each state j there exists a positive integer m_j such that $P_{j,j}^{(m)} > 0$ for all $m \geq m_j$.

Let $K = \max_{i,j} \{n_{i,j} + m_j\}$. Then $K - n_{i,j} \geq m_j$ for every pair i, j , and

$$P_{i,j}^{(K)} \geq P_{i,j}^{(n_{i,j})} P_{j,j}^{(K-n_{i,j})} > 0.$$

Thus every entry of $\mathbf{P}^K$ is strictly positive, so the chain is regular.

Problem 1.5. (Exam 2023) Consider the Markov chain $\{X_n : n = 0, 1, \dots\}$ with state space $\{0, 1, 2\}$, and transition probability matrix

$$\mathbf{P} = \begin{bmatrix} \alpha & 1-\alpha & 0 \\ 0 & \beta & 1-\beta \\ 1-\gamma & 0 & \gamma \end{bmatrix}$$

for some known $\alpha, \beta, \gamma \in [0, 1]$.

- What conditions are necessary for the Markov chain to be irreducible?
- Assume the irreducibility condition holds:
 - What are the equivalence classes? What are the periodicities of each state, and under what conditions?
 - Is the Markov chain positive recurrent? Does the chain have a stationary distribution? If so, is it a limiting distribution?

Solution.

- The chain is irreducible if and only if $\alpha, \beta, \gamma < 1$.
- If the chain is irreducible, all states communicate with each other, and there is only one equivalence class $\{0, 1, 2\}$.

When $\alpha, \beta, \gamma = 0$, we have

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix},$$

and it is clear that $d(0) = d(1) = d(2) = 3$. Otherwise, we can have self-loops in one or more states, and the chain is aperiodic with $d(0) = d(1) = d(2) = 1$.

- As the chain is finite and irreducible, it is positive recurrent. It therefore also has a unique stationary distribution. If the chain is also aperiodic, this stationary distribution is also the limiting distribution.

Problem 1.6. We throw a fair coin repeatedly, and denote each outcome by H (heads) or T (tails). Let n denote the n -th throw in a sequence, and define a stochastic process $\{X_n : n = 0, 1, \dots\}$ by

$$X_n = \text{The number of consecutive throws of H, including throw } n, \quad n = 0, 1, \dots$$

Do the following:

- Explain why this is a Markov process. Specify the state space, and find the transition probabilities.
- Is this Markov chain irreducible?
- For each state, calculate its period.
- Show that the states are recurrent.
- Calculate the long-run proportion of time that the sequence of throws ends in three or more heads.

Solution.

- Each throw is independent of the previous throws. Only the previous state affects the probability of transitioning to a new state. The transition probabilities are

$$P_{i,0} = 1/2, \quad P_{i,i+1} = 1/2, \quad i = 0, 1, \dots,$$

and the state space is $\{0, 1, \dots\}$.

- We can reach 0 from any state, and we can reach any state from 0 by repeatedly getting heads. Therefore, all states communicate, and the Markov chain is irreducible.
- Here it is easy to make a mistake and think that $d(i) = i + 1$ for all states i . I provide two explanations below, with and without the use of [Theorem 1.8](#):
 - From [Theorem 1.8](#) we know that all states have the same period since they communicate. Clearly $d(0) = 1$ since we can throw tails and return to 0 in one step. Therefore, $d(i) = 1$ for all states i .
 - We know $d(0) = 1$, but now consider some state $i > 0$. One way to return to i is in $i + 1$ steps by throwing a tail and then i heads. However, we can also throw two tails and then i heads, which takes $i + 2$ steps. Therefore, $d(i) = \gcd(i + 1, i + 2) = 1$.

For both explanations, we conclude that all states are aperiodic.

- As all states communicate, they are either all recurrent or all transient by [Theorem 1.11](#). We consider state 0, where

$$f_{0,0}^{(n)} = 1/2^n, \quad n = 1, 2, \dots$$

and therefore

$$f_{0,0} = \sum_{n=1}^{\infty} 1/2^n = 1$$

by the geometric series. Since 0 is recurrent, all states are recurrent.

- Can also show positive recurrence. The mean recurrence time of state 0 is

$$m_0 = \sum_{n=1}^{\infty} n f_{0,0}^{(n)} = \sum_{n=1}^{\infty} n (1/2)^n = 2 < \infty.$$

As positive recurrence is a class property, all states are positive recurrent. As we previously showed that the chain is irreducible, we can use [Theorem 1.16](#) to find the stationary distribution. Let π_0 denote the stationary distribution of state 0, then

$$\pi_0 = 1/m_0 = 1/2.$$

To find the stationary distribution for the other states, we see

$$\pi_1 = \sum_{k=0}^{\infty} \pi_k P_{k,1} = \pi_0 P_{0,1} = 1/4,$$

$$\pi_2 = \sum_{k=0}^{\infty} \pi_k P_{k,2} = \pi_1 P_{1,2} = 1/8,$$

and so on. By the ergodic theorem for irreducible positive recurrent Markov chains, the long-run proportion of time that the sequence of throws ends in three or more heads is therefore

$$\sum_{k=3}^{\infty} \pi_k = 1 - \pi_0 - \pi_1 - \pi_2 = 1/8.$$